Massey University
ALBANY CAMPUS

EXAMINATION FOR
159.302 ARTIFICIAL INTELLIGENCE
Semester One - 2014

Time Allowed: THREE (3) Hours

INSTRUCTIONS

Answer ALL five (5) questions.

This final examination contributes 60% to the final assessment.

Calculators are permitted.

QUESTION 1: STATE-SPACE SEARCH [19 marks total]

A robot has just made a startling discovery, while on a mission in Mars. It has found a cure for cancer in the form of an alien mineral. However, its battery is almost running out and its right wheel is damaged. In order to recharge and get immediate repair, it has to find its way back to Napoleon's rocket ship.

The robot has to minimise energy consumption wisely, while navigating around obstacles, towards its destination.

The following cases need to be taken into account in the path calculations:

- Turning 90° **right** at a junction takes time equivalent to travelling a path of length 2 (extra mechanical parts are involved in a turning action, as compared to moving straight).

- Turning 90° **left** at a junction takes time equivalent to travelling a path of length 3 (unfortunately, the robot's right wheel needs to do more work to turn the robot to the left, but the right wheel is slightly damaged; therefore, it requires more energy).

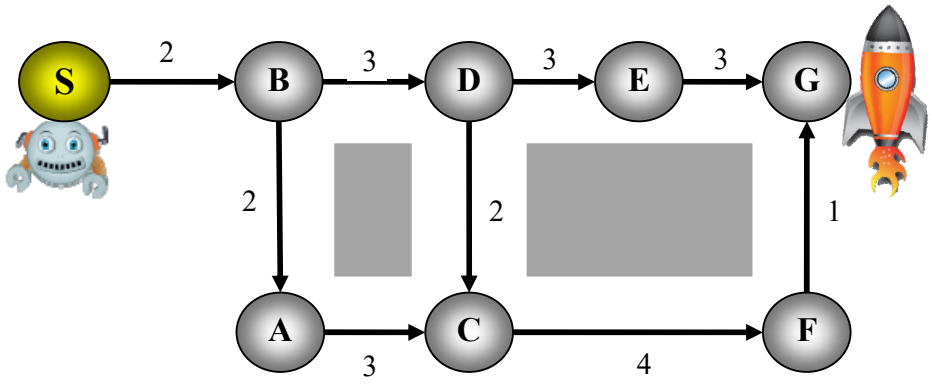
All path lengths are shown in the graph. Obstacles are indicated as regions coloured grey.

Hint: For example, from B to A, the total energy cost is equal to
= 2 length units (for making a right turn) + 2 left units (actual distance indicated in the graph)
= 4 length units.

The heuristic values, associated with each node are as follows:
S = 5, A= 3, B = 2, C=4, D=1, E=2, F=3, G=0.

The objective is to calculate the path that will require the <u>least amount of energy</u> (not the shortest distance travelled), from position **S** to the rocket ship (marked as node **G**).

Break ties alphabetically.



1.1. **Search Tree**

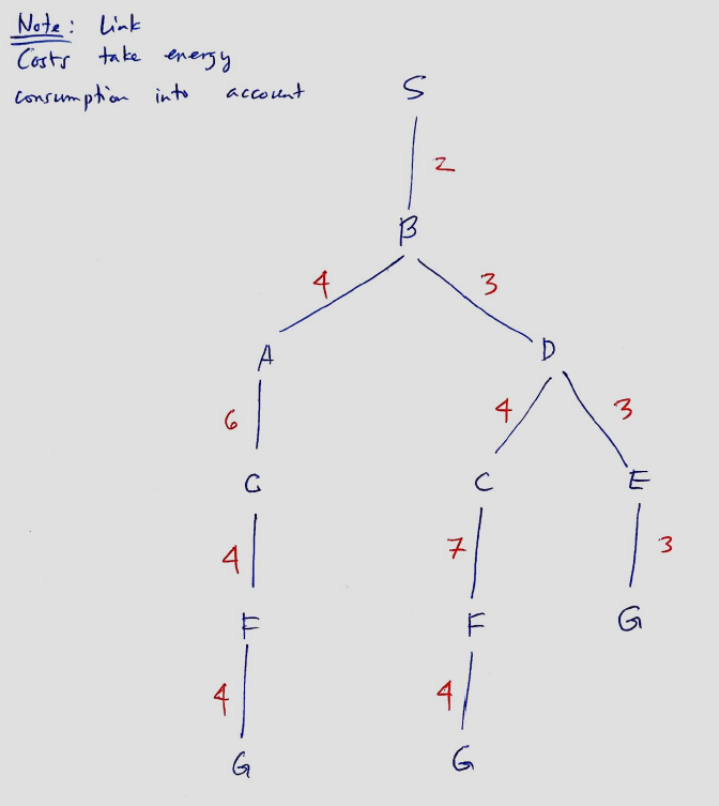
Draw the search tree, ordering the descendants of each node alphabetically from left to right.

[**2** marks]

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Solution:

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Simulation of Search Algorithms

General Instructions (for Parts 1.2-1.4):

In your final answer, write the sequence of **states expanded** and the **path found** to the rocket ship by the specified search methods. Indicate the state name for each step in the sequence, including **S** and **G**.

For each search method simulation, use the following table format to show your complete solution.

Step	Dequeued	Enqueued	Visited List or Expanded List
0		(S)	
1			
…and so on			

Sequence of State Expansion: _____
Path found to the rocket ship: _____

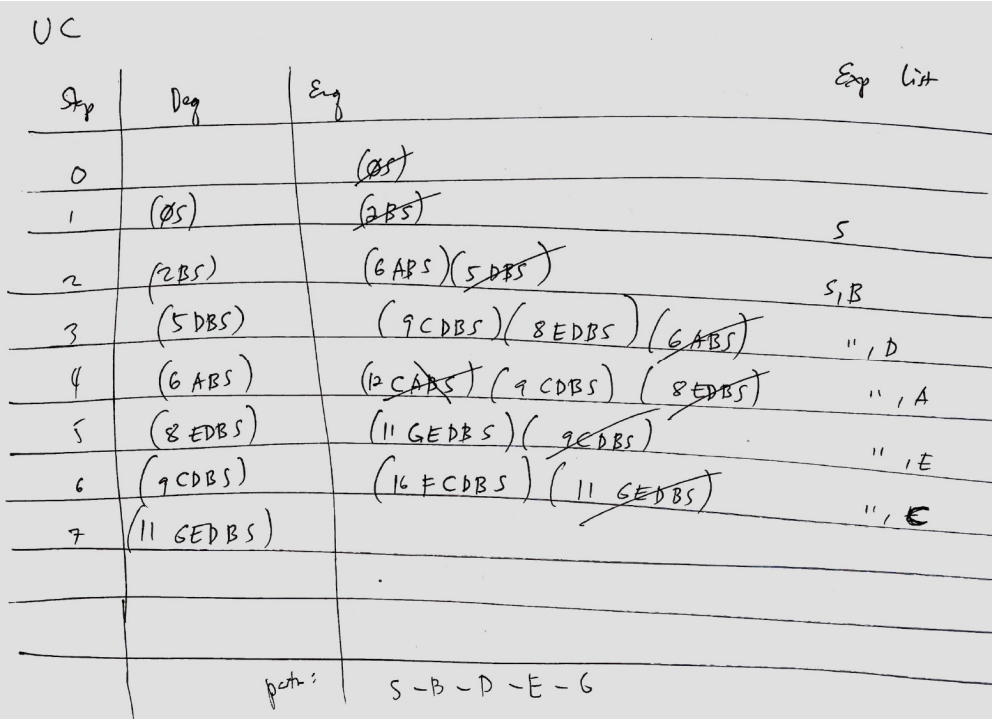
1.2. Depth-First Search without using the Visited List.

[5 marks]

1.3. Uniform-Cost Search using the Strict Expanded List. You should fill-in the Expanded List column in the table.

[5 marks]

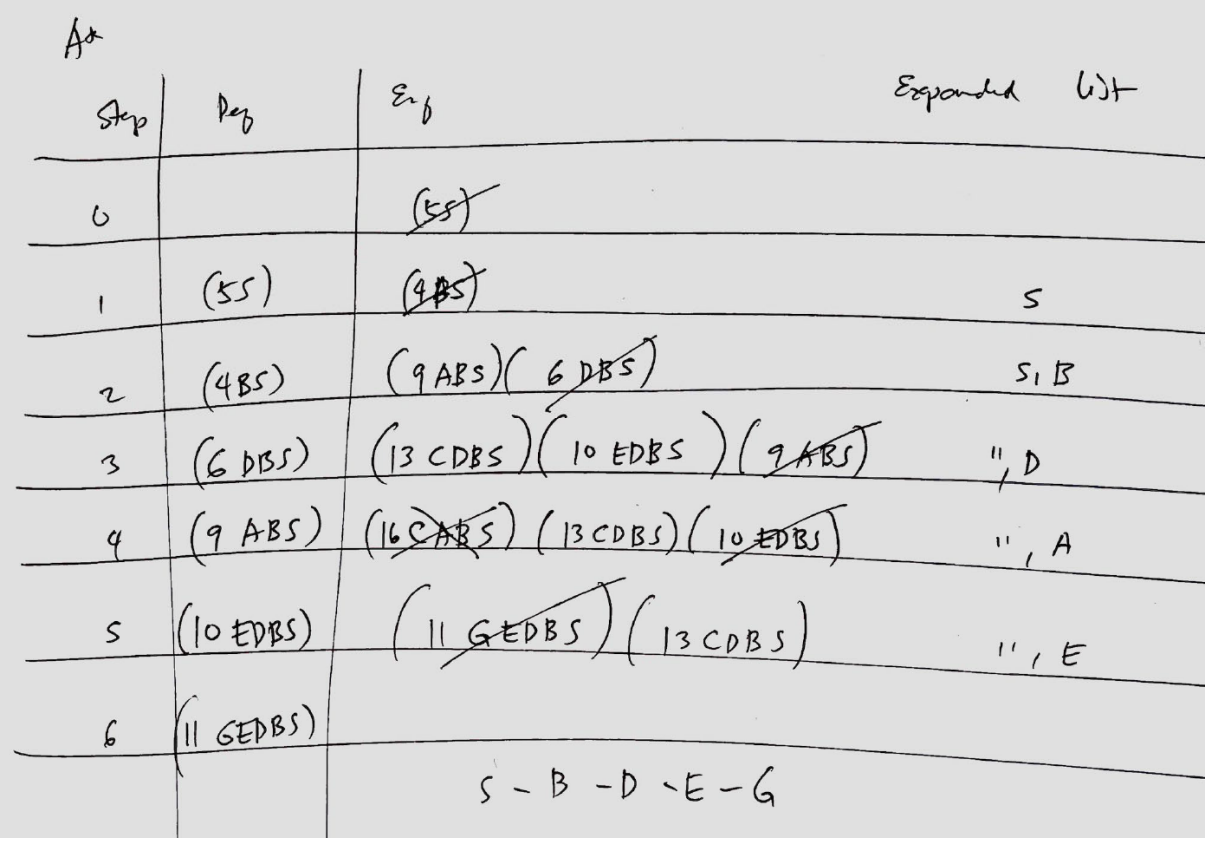
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Solution:
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1.4. A* using the Strict Expanded List. You should fill-in the Expanded List column in the table.

[5 marks]

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Solution:
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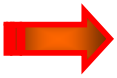
1.5. **Heuristics**

Calculate the Sum of Manhattan distance heuristic for the given Start state of the 8-Puzzle problem.

8-Puzzle

6	2	8
	3	5
4	7	1

Start



	1	2
3	4	5
6	7	8

Goal

[2 marks]

QUESTION 2. CONSTRAINT SATISFACTION PROBLEM [16 marks total]

Consider assigning colours to a checkerboard so that squares that are adjacent vertically or horizontally do not have the same colour.

1	2	3
4	5	

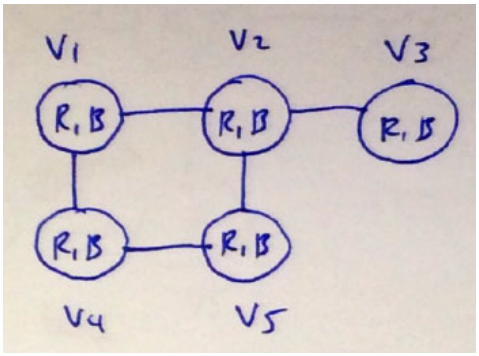
Construct a CSP formulation of the problem according to the following requirements:

- Let the squares be the variables and the colours be the values.
- Limit the possible colour values to only red (R) and black (B).
- Set the initial domains of all variables equal to {R, B}.

2.1. Draw the constraint graph representation of the problem.

[2 marks]

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Solution:
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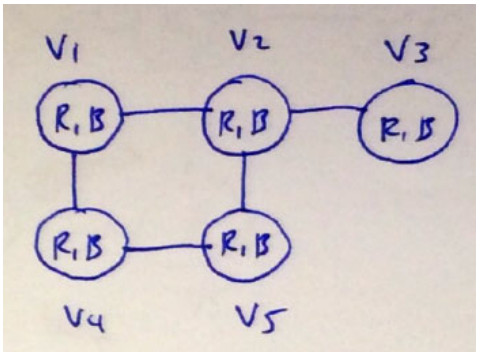
For each of the next succeeding CSP algorithm simulation questions (items 2.2 – 2.5), use the following initialisation values:

- Set the initial domain of variable 5 equal to {B}.
- Set the initial domains of all the other variables equal to {R, B}.

2.2. Apply the full constraint propagation algorithm. Show your answer using a constraint graph.

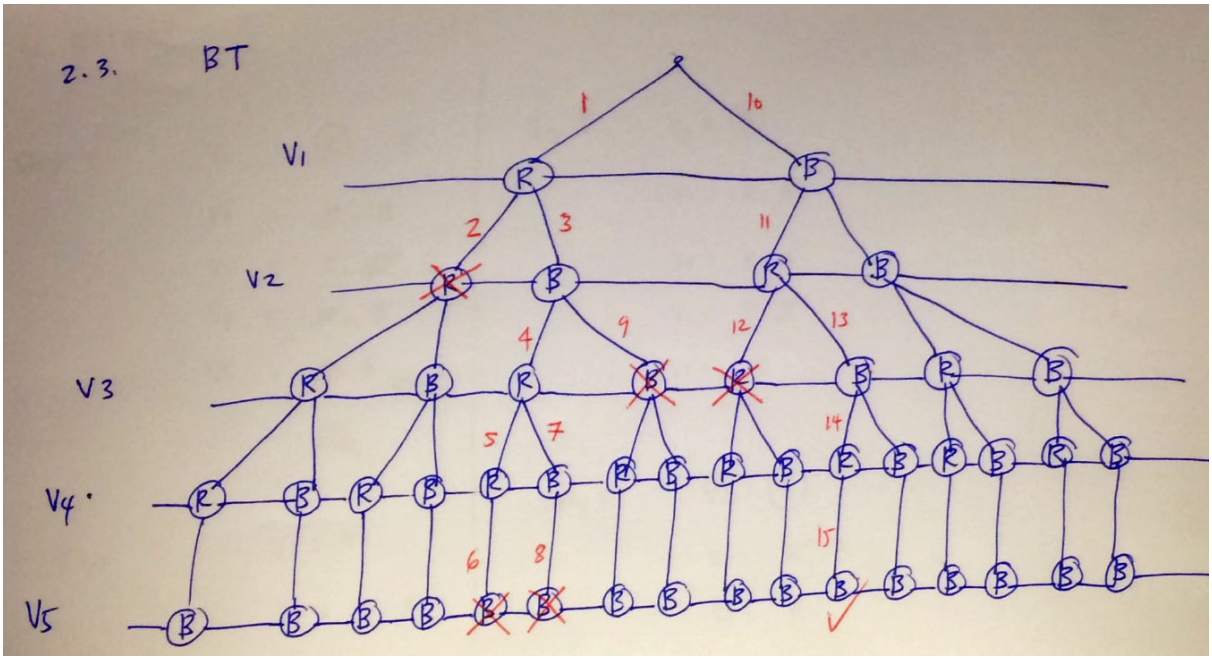
[2 marks]

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Solution:
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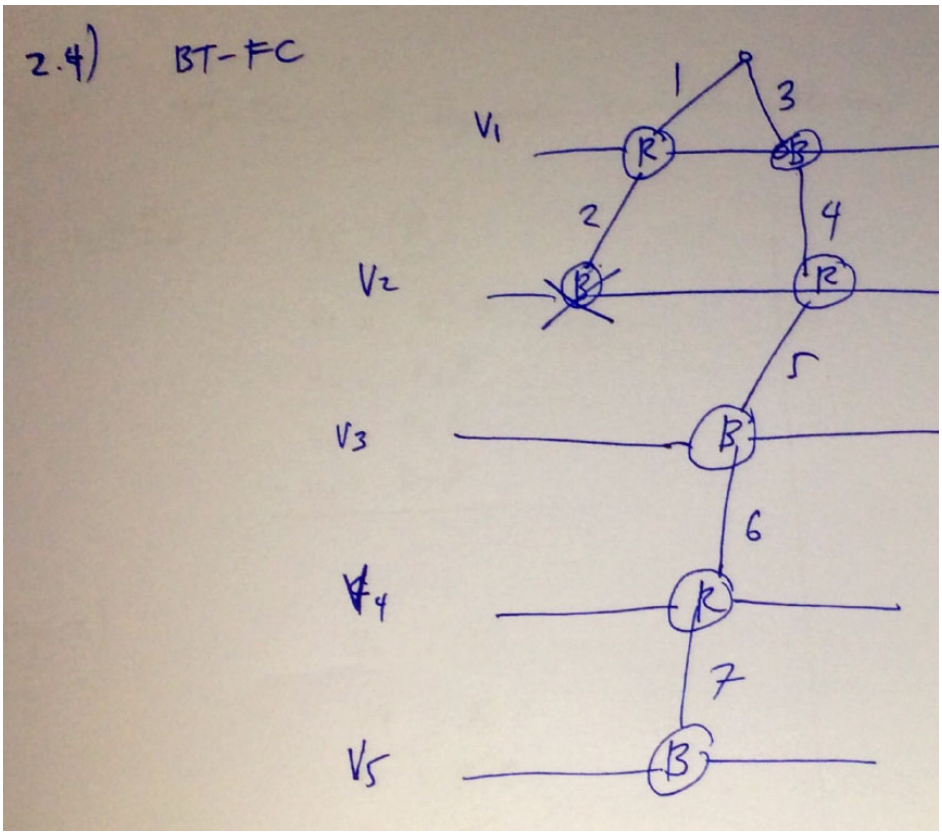
2.3. Apply the pure backtracking algorithm. Show your answer using a constraint graph.
[3 marks]

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Solution:
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2.4. Apply the backtracking with forward checking algorithm. Show your answer using a constraint graph.
[4 marks]

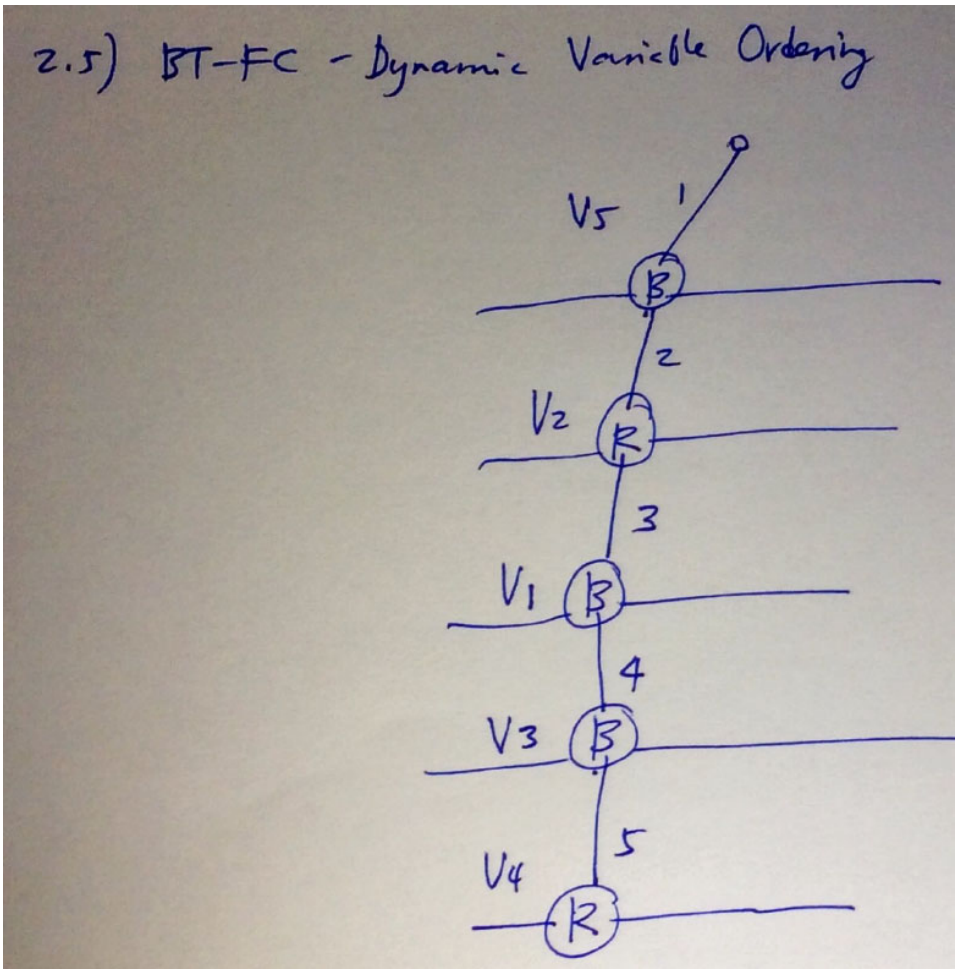
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Solution:
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2.5. Apply the backtracking with forward checking, with dynamic variable ordering algorithm. Show your answer using a constraint graph.

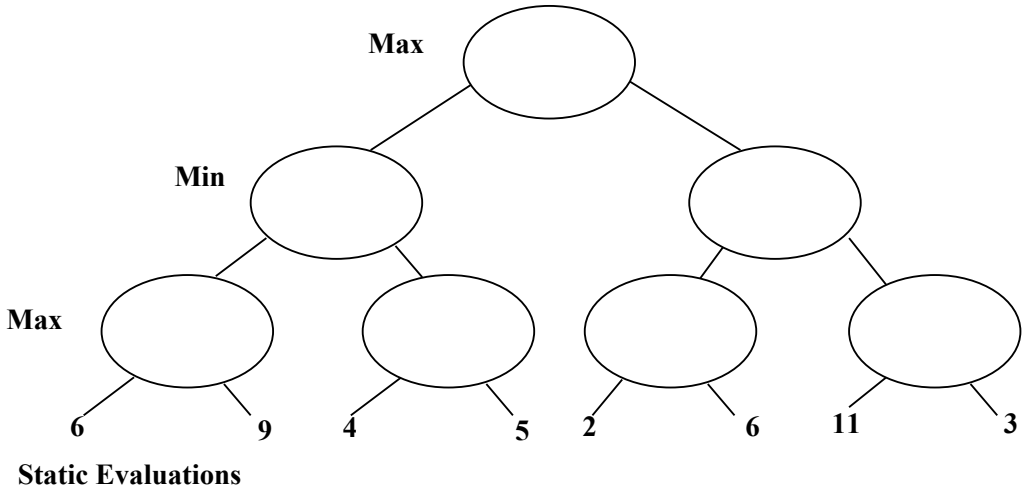
[5 marks]

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Solution:
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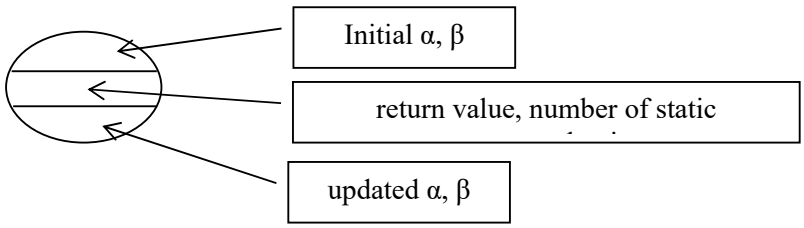
QUESTION 3. GAMES: Min-Max and Alpha-Beta Pruning [8 marks total]

3.1. Simulate the **MinMax** algorithm on the Game Tree provided below. Show your complete answer by filling the interior of all the nodes with the **return value** and the **number of static evaluations**. Copy your answer (complete game tree) on the blue answer book..

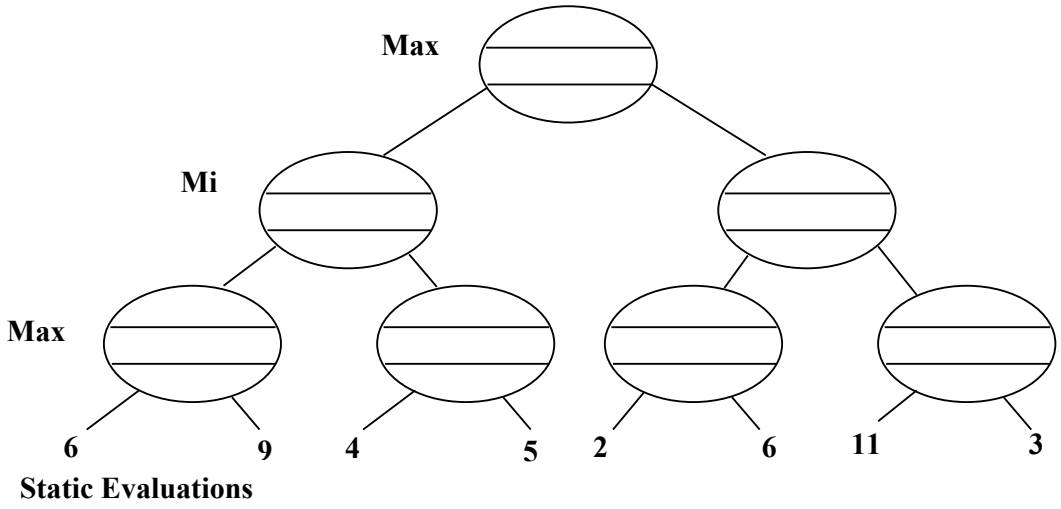


[**2 MARKS**]

3.2. Simulate **Alpha-Beta Pruning** (from left to right) on the game tree provided below. Show your complete answer by filling each of the three sections of the interior of the nodes with the required information. Copy your answer (complete game tree) on the blue answer book.



Game Tree:



[6 MARKS]

QUESTION 4: Fuzzy Logic [7 marks total]

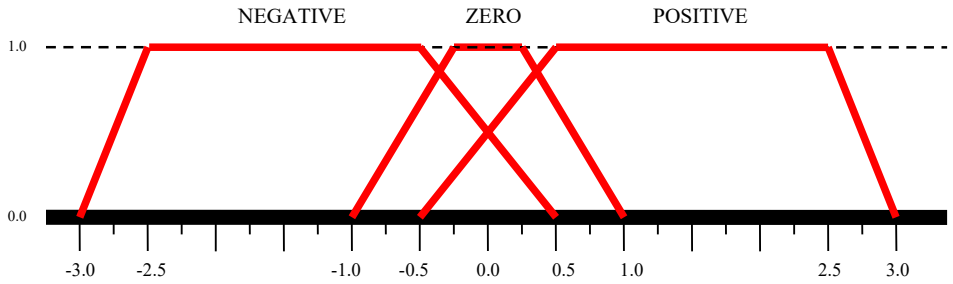
Consider the following **FAMM** with two inputs: **x** and **y**

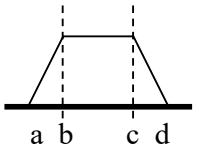
Outputs:
NL= -3
NS= -2.5
ZE= 0
PS= 2.5
PL= 3.0

y \ x	N	ZE	P
N	NL	NS	NS
ZE	NS	ZE	PS
P	PS	PS	PL

Fuzzy Sets = { Negative (N), Zero (ZE), Positive (P) }

Assume that the following Fuzzy Sets (implemented using Trapezoidal membership functions) are applicable to inputs **X** and **Y**.





For all the questions below, refer to the given FAMM above and the corresponding numerical equivalent of the output terms.

4.1.1. Calculate the degree of membership of an input **X= -0.25** to the Fuzzy set **N**.
[**1 mark**]

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Solution:
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x	-0.25
a	-3
b	-2.5
c	-0.5
d	0.5
$F_{Neg}(x)$	0.75

4.1.2. Calculate the degree of membership of an input **Y= 0.25** to the Fuzzy set **P**.
[**1 mark**]

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Solution:
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y	0.25
a	-0.5
b	0.5
c	2.5
d	3
$F_{Pos}(x)$	0.75

4.2. Given the inputs in item 5.1, calculate the degree of firing (weight) for rule #3 (i.e. W3) using Zadeh's fuzzy AND as a logical connective.

		x		
		N	ZE	P
y	N	W1	W4	W7
	ZE	W2	W5	W8
	P	**W3**	W6	W9

[**2 marks**]

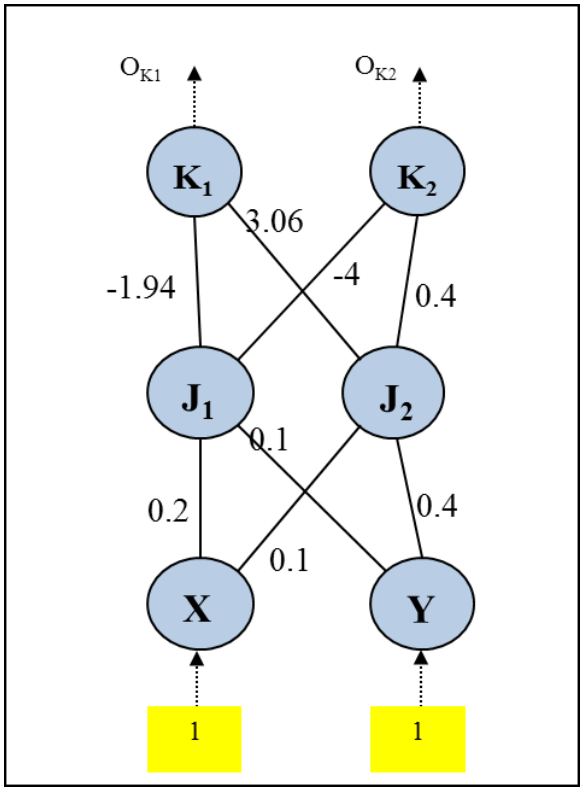
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Solution:
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Degree of firing (weight) for rule #3 = min(0.75, 0.75) = 0.75

4.3. Defuzzification: Write the complete formula for calculating the centre of mass for this particular problem in terms of the weights (e.g. W1, W2, etc.) and the given outputs in linguistic terms (e.g. NL, NS, PS, etc.).

[**3 marks**]

QUESTION 5. BACKPROPAGATION LEARNING [10 marks total]



Refer to the figure depicting a multi-layer feed-forward network. The activation and error values of the output nodes K_1 and K_2 have been calculated already based on the given training pattern, and that the weights W_{j1k1}, W_{j1k2}, W_{j2k1}, W_{j2k2} have been updated already (the adjusted weights are shown in the figure. Furthermore, we have the following information about the given neural network:

Learning rate $\eta = 2$
No bias nodes are used.
activation value of k1: $O_{k1} = 0.67$,
target output value of k1: $t_{k1} = 0.91$
activation value of k2: $O_{k2} = 0.11$
target output value of k2: $t_{k2} = 0.01$
error signal on output node k1: $\delta_{k1} = 0.05$
error signal on output node k2: $\delta_{k2} = 0.00$
activation value of node j1: $O_{j1} = 0.57$

training pattern:

Input: x	Input: y	Target output: T_{K1}	Target output: T_{K2}
1	1	0.91	0.1

Apply the given values and refer to the figure to solve for the following:

5.1. **Error signal on hidden node J1**: (δ_{j1})
Write the general formula first, and then substitute the given values to find the error value.

[**4 marks**]

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Solution:
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Error signal on hidden node J1: (δj1) = -0.02

5.2. **Next weight update on W_{xj1}**
Write the general formula first, and then substitute the given values to find the new value for the weight.

[**4 marks**]

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Solution:
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Next weight update on Wxj1 = 0.15

Error Calculations

Refer to the table below. Given a neural network with 3 output nodes and a table detailing the network's response on 3 training patterns, calculate the following error value:

5.3. Sum of Squared Errors [**2 marks**]

Training Pattern 1

output nodes	1	2	3
T_k	0.7	0.7	0.7
O_k	0.4	0.5	0.8
$(T_k - O_k)^2$			

Training Pattern 2

output nodes	1	2	3
T_k	0.9	0.9	0.9
O_k	0.8	0.5	0.6
$(T_k - O_k)^2$			

Training Pattern 3

output nodes	1	2	3
T_k	0.1	0.1	0.1
O_k	0.4	0.2	0.3



T_k – target output for output node k
O_k – actual output of output node k

////////////////// **Solution:** //////////////////

Training Pattern 1				
output nodes	1	2	3	sum
tk	0.7	0.7	0.7	
ok	0.4	0.5	0.8	
abs(tk-ok)	0.3	0.2	0.1	0.6
(tk-ok)^2	0.09	0.04	0.01	0.14

Training Pattern 2				
output nodes	1	2	3	sum
tk	0.9	0.9	0.9	
ok	0.8	0.5	0.6	
abs(tk-ok)	0.1	0.4	0.3	0.8
(tk-ok)^2	0.01	0.16	0.09	0.26

Training Pattern 3				
output nodes	1	2	3	sum
tk	0.1	0.1	0.1	
ok	0.4	0.2	0.3	
abs(tk-ok)	0.3	0.1	0.2	0.6
(tk-ok)^2	0.09	0.01	0.04	0.14

SSE = 0.27

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